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# FINITE ELEMENT MODELING OF ELECTROCHEMICAL IMPEDANCE SPECTROSCOPY IN ALL-SOLID-STATE BATTERIES USING FENICSx

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All-solid-state thin-film batteries are key components for the development of autonomous integrated systems, where performance depends on complex charge transport and interface kinetics [1]. Electrochemical Impedance Spectroscopy (EIS) represents a non-destructive technique used to characterize these phenomena by decoupling the resistive and capacitive contributions of the solid electrolyte and its interfaces [2]. In this work, we implement a numerical framework within FEniCSx to simulate the EIS response of such systems based on the Poisson-Nernst-Planck (PNP) model. This model couples electrostatic potential and ionic concentrations, accounting for the formation of the electrical double layers at the electrode-electrolyte interfaces [3]. We utilize the Unified Form Language (UFL) to perform an automatic linearization of the non-linear coupled system around various electrode bias potentials. This approach allows for a direct resolution in the frequency domain. The central feature of the implementation is the post-processing of the frequency-domain results, using a residual-based current calculation to ensure strict electrical conservation. We demonstrate the framework's validity by comparing simulated Nyquist plots with experimental data obtained from thin -film batteries. This work provides a physically consistent tool for the study of electrochemical phenomena in solid-state energy storage.

## References

- [1] Wu, T., Dai, W., Ke, M., Huang, Q., & Lu, L. (2021). All-Solid-State Thin Film  $\mu$ -Batteries for Microelectronics. *Advanced Science*, 8(19), 2100774. DOI: 10.1002/advs.202100774
- [2] Orazem, M. E., & Tribollet, B. (2020). A tutorial on electrochemical impedance spectroscopy. *ChemTexts*, 6(2), 12. DOI: 10.1007/s40828-020-0110-7
- [3] Bazant, M. Z., Thornton, K., & Ajdari, A. (2004). Diffuse-charge dynamics in electrochemical systems. *Physical Review E*, 70(2), 021506. DOI: 10.1103/PhysRevE.70.021506